Final Paper

STOR 320.01 Group 14
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Introduction

Diabetes is now the 8th-leading cause of death in the United States and its prevalence continues to rise, but most screening algorithms rely on laboratory tests that do not arrive until metabolic damage is already in progress (Centers for Disease Control and Prevention [CDC], 2024). The growing prevalence of diabetes poses a major public health challenge, affecting over 38.4 million Americans, with another 8.7 million adults potentially living undiagnosed. Ranked as the eighth leading cause of death in the United States, diabetes is a chronic condition characterized by the body's inability to properly regulate blood glucose (CDC, 2024). While clinical advancements and updated guidelines from the American Diabetes Association (ADA) continue to improve care , disparities rooted in social and behavioral factors persist (American Diabetes Association, 2024). Beyond objective medical tests, an increasing focus has been placed on subjective, self-reported data, highlighting the importance of lifestyle choices and individual health behaviors in diabetes outcomes (CDC, 2024).

Phone-based surveillance systems like the CDC's Behavioral Risk Factor Surveillance System (BRFSS) gather low-cost, self-reported information on over 400,000 adults annually, presenting an opportunity to identify risk earlier and at scale (CDC, 2023). Extending that concept, we ask two questions: (1) Can a person's diabetes status be predicted from behavioural and social-determinant responses alone? (Hill-Briggs et al., 2020) (2) Among dozens of candidate indicators, which ones carry the greatest predictive weight and therefore merit the limited attention of clinicians, insurers, and public-health planners? In answering Question 1, a data owner obtains a low-cost triage tool. This could provide a valuable early-warning system that enables targeted prevention strategies and empowers individuals to take proactive steps toward their health (Berkowitz et al., 2015). For healthcare providers and policymakers, the ability to intervene earlier based on behavior rather than waiting for symptoms to worsen could reduce long-term costs and improve population health outcomes (Magnan, 2021). Nevertheless, answering Question 2, by identifying the strongest predictors, whether physiological (such as BMI or blood pressure), behavioral (such as exercise or diet), or demographic (such as income, age, or ethnicity), we can help data owners prioritize the most impactful variables in screening models and public health messaging (CDC, 2023). Understanding which factors carry the most weight not only refines diagnosis but can also inform more efficient, data-driven prevention efforts (Hill-Briggs et al., 2020).

Our study also yields depth into an integrative and innovative effort in early prediction health dataset. Our study compares multinomial logistic regression, elastic-net regularization, and pruned decision trees on BRFSS survey data to predict diabetes risk, while intentionally showing why k-nearest neighbors fails. The innovation and creativity of this choice shows through the cross-model interpretation, because each model addresses a different real-world challenge: logistic regression offers interpretable risk odds, elastic-net handles collinearity among social determinants, and decision trees capture nonlinear health patterns. All together, by evaluating not just what works but why certain models fail, our project provides a practical, transparent roadmap for using everyday health data to better predict diabetes. Data scientists

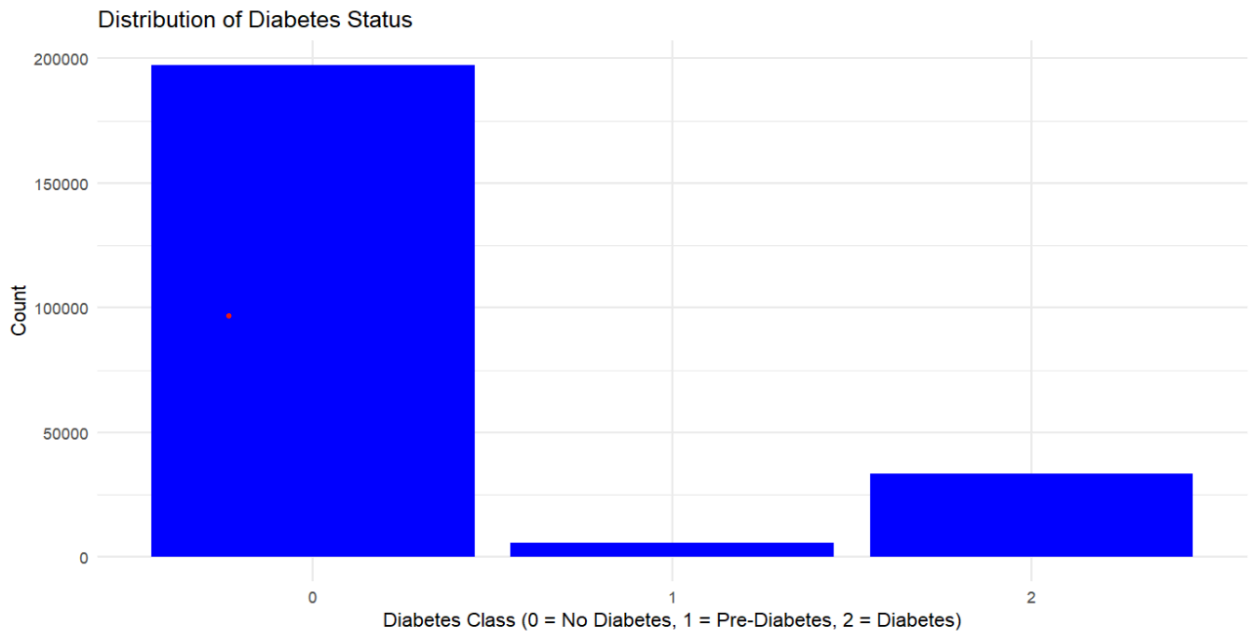
gain a clear comparison of model trade-offs for imbalanced, high-dimensional health datasets, while public health leaders gain insights into how low-cost surveys can power early interventions.

Together, these questions and models explored aim to shift the narrative around diabetes from reactive treatment to proactive insight. By leveraging both behavioral and biomedical data, we hope to identify patterns that can lead to earlier diagnoses, reduce disparities, and personalize care. As diabetes continues to rise globally, answering these questions holds the potential for transformative change in clinical settings, in communities, and in the lives of individuals navigating this disease.

Data

Our dataset, ***diabetes_012_health_indicators_BRFSS2021*** contains participant responses collected from a 2021 telephone survey conducted by the Behavioral Risk Factor Surveillance System (BRFSS) & CDC pertaining to 236,378 individuals. The data is a cleaned and consolidated dataset created from the BRFSS 2021 dataset found on Kaggle obtained by Julian Nazreen. The two types of interviews conducted are through landline telephones, which include random selection among adults within households, and cellular telephone which are conducted with adults who are contacted on personal (nonbusiness) cellular telephones. The survey process included questioning participants on demographic characteristics, plus queries on current health behaviors. Additionally, variables were calculated from responses to questions in the survey, allowing data to be stratified and weighted to derive contingent variables from question responses to calculate risk factor, and categorize and classify respondents. Since the initial inclusion of cell phone respondents, the proportion of the sample that is interviewed by cell phone has increased. In many states, individuals reached via cell phone now make up the majority of the sample. Although telephone surveys offer valuable data, their effectiveness can be limited by several challenges, including declining response rates, concerns about the accuracy of self-reported information, the quality of respondent engagement, and the potential for both nonresponse and measurement bias.

The target variable, ***Diabetes_012***, has 3 classifications represented by 0,1, 2 to organize participants with no diabetes or only during pregnancy, prediabetes, and diabetes respectively. This data represents an asymmetric, right-skewed distribution favoring classifier 0 of 197,191 individuals while 5,619 participants represent prediabetes (1) and 33,568 individuals who have diabetes (2).



Our data includes ***22 variables***, divided by ***subjective*** and ***objective*** variables:

Objective Variables:

- ***HighBP***: 0 = no high BP, 1 = high BP
- ***HighChol***: 0 = no high cholesterol, 1 = high cholesterol
- ***CholCheck***: 0 = no cholesterol check in 5 years, 1 = yes cholesterol check in 5 years
- ***BMI***: Body Mass Index (Weight (in pounds) / [Height (in inches)]^2 x 703))
- ***Stroke***: (Ever told) you had a stroke. 0 = no, 1 = yes
- ***HeartDiseaseorAttack***: Coronary Heart Disease (CHD) or Myocardial Infarction (MI) 0 = no, 1 = yes
- ***Age***: 13-level age category: scale 1-13: 1 = 18-24, 2 = 25-29, 3 = 30-34, 4 = 35-39, 5 = 40-44, 6 = 45-49, 7 = 50-54, 8 = 55-59, 9 = 60-64, 10 = 65-69, 11 = 70-74, 12 = 75-79, & 13 = 80 or older
- ***Education***: Education level: scale 1-6: 1 = Never attended school or only kindergarten, 2 = Grades 1 through 8 (Elementary), 3 = Grades 9 through 11 (Some high school), 4 = Grade 12 or GED (High school graduate), 5 = College 1 year to 3 years (Some college or technical school), & 6 = College 4 years or more (College graduate)
- ***Income***: scale 1-8: 1 = less than $10,000, 2 = less than $15,000, 3 = less than $20,000, 4 = less than $25,000, 5 = less than $35,000, 6 = less than $50,000, 7 = less than $75,000, & 8 = $75,000 or more

Adjustments to Scaling:

BMI was adjusted using Z-score scaling. **Age** was adjusted using scale() or Z-score based on continuous, right skewed data. **Income** was adjusted using Log + Scale or (0-1) with ordinal encoding. **Education** was redefined as an ordinal, linear variable. The rest of the objective variables were left as continuous (0-1).

Subjective Variables:

- ***PhysActivity***: Physical activity in past 30 days - not including job 0 = no, 1 = yes
- ***Fruits***: Consume fruit 1 or more per day 0 = no, 1 = yes
- ***Veggies***: Consume vegetables 1 or more per day 0 = no, 1 = yes
- ***HvyAlcoholConsump***: Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per
- ***AnyHealthcare***: Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no, 1 = yes
- ***NoDocbcCost***: Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no, 1 = yes
- ***GenHlth***: Would you say that in general your health is: scale 1-5: 1 = excellent, 2 = very good, 3 = good, 4 = fair, 5 = poor
- ***MentHlth***: Now thinking about your mental health, which includes stress, depression, and problems with emotions during the past 30 days (0-30 days).
- ***PhysHlth***: Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days (0-30 days).
- ***DiffWalk***: Do you have serious difficulty walking or climbing stairs? 0 = no, 1 = yes
- ***Sex***: 0 = female, 1 = male
- ***Smoker***: Have you smoked at least 100 cigarettes in your entire life? [Note: 5 packs = 100 cigarettes] 0 = no, 1 = yes

Adjustments to Scaling:

GenHlth adjustments based on the scale (1-5) were redefined as ordered categorical variables. A linear format of **GenHlth** performs worse for AIC and accuracy, since **GenHlth** is ordinal making it ordered is best to use for meaningful and appropriate data. **MentHlth** and **PhysHlth** were defined as categorical, based on number of days (1-30): log1p() + scale() for right-skewed count. The rest of the subjective variables are left as continuous (0-1).

Diabetes_012	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke	HeartDiseaseorAttack	PhysActivity	Fruits	Veggies	HvyAlcoholConsu
0	0	1	1	15	1	0	0	0	1	1	
2	1	0	1	28	0	0	1	0	1	0	
2	1	1	1	33	0	0	0	1	1	1	
2	0	1	1	29	0	1	1	1	1	1	
0	0	0	1	24	1	0	0	0	0	0	
0	1	0	1	40	1	0	0	1	1	1	
0	1	1	1	27	1	0	0	0	0	1	
2	0	0	1	24	0	0	1	0	0	1	
0	1	1	1	30	0	0	0	0	1	1	
0	1	1	1	36	1	0	0	0	0	0	

Results

Part I:

Using a multinomial logistic regression, we can accommodate nominal and ordinal categorical data, as well as scaled continuous variables within our dataset. This model also helps predict our dependent variable by handling more than two categories, providing flexibility to manipulate and detail to interpret variables' influence on overall predictability, combined with the outcomes of log odds to be modeled as linear combinations of the predictor variables. We began by modeling an empty model and baseline multinomial logistic regression without any scaling as the initial benchmark of our data to identify the signal of each variable in association with diabetes status. Splitting our model based on subjective and objective variables reveals the fragility of our data, based on the bare representation of Classes 1 and 2, with both models veering in favor of predicting Class 0**.** While the objective model performs better overall due to its higher accuracy, lower AIC, and a slight increase in sensitivity with Class 2, our focus on the subjective data remains in stabilizing class imbalance and dilating predictive powers.

By upsampling our data, we contrive duplications of Class 1 and 2 observations to detect an equal representation of all three classes while making finer predictions based on the Kappa value; our model finally predicts prediabetes and diabetes. This balance is provided with the caveat of dropping accuracy by nearly 30% while accepting the loss of accuracy for the insight provided by additional representation. Due to the highly correlated predictors and focus on identifying a subset of important variables, using an Elastic Net Model (balancing Lasso and Ridge) with upsampling and use of class weights moved forward as a regularization technique while being weary of eliminating and refitting variables that change other predictions of coefficients. *Lasso (L1)* removes and shrinks coefficients to zero for unsuitable or redundant features to improve overfitting based on a proportion similar to the sum of absolute values of coefficients. *Ridge (L2)* controls coefficient size by creating a penalty term proportional to the ordinary least square objective function. By discouraging larger coefficient values, the variance of parameter estimates adjusts precision for suitable conditions susceptible to more minor changes. *Class 1* persists in the challenge of proper representation, performing a tiny percentage higher with upsampling by a range of 29-30%. However, the slight negligence in the difference between sensitivity and the *Kappa* value is short-cut by the advantages of class weights providing a higher accuracy and *Class 2* performance,

resourceful to maintaining the overall harmony and balance of the dataset. By utilizing cross-validation on our upsampled elastic net model, the prevention of overfitting discourages complexity and reduces any sustained noise captured by the training model. While *Class 1 and 2* representation is attuned with *Class 0* now, distinguishing each class remains challenging with our predictors. Lambda's relatively low penalty strength can help us understand the weight of these variables.

Based on the tuning of *L1 and L2* penalties, adjustments to our alpha value were determined by a 5-fold cross-validation method to tune the value of *alpha* and *lambda*. The performance of our five folds averaged to get an estimation for the optimal set of parameters. By testing lambda variations, we obtained the optimal value for minimizing deviance and balancing our model performance. This decreased our accuracy by 10% and continued to demonstrate the strength of *Class 0* obstructs the prevalence of *Class 1 and 2*, while 15 of the 22 predictors remained non-zero through cross-validation. Through the log-odds ratio, we could see that our model automatically selects coefficients that are not to be included (equal to 0). The caveat of a good model is to recognize that the model's score does not correlate to its probability, even if assigning higher probabilities to positive cases and lower probabilities to negative cases while the predictor score increases. This led to adjusting our focus on identifying variables to detect *Class 1* and delegating the strength of coefficients to the best predictors. The log-odds ratio indicated that **AnyHealthcare** and **NoDocCost** were our two most significant predictors, while **MentHlth**, **Income**, and **Education** had slightly smaller effects. While the impact of these predictors is important to understand, they can't suggest prediabetes, especially when interaction terms are applied. This model highlighted the extensive bias of *Class 0*, as no pairwise interaction improved the model's ability even with the strength of the variables. By adjusting alpha values in association with cross-validation on our interaction terms, we achieved the best deviance when **α=1** while maintaining an overall 63% accuracy but decreasing *Class 1* sensitivity. Pairwise interactions boosted the number of terms (11 choose 2) to increase accuracy, *Class 1* sensitivity, and retrieve the best lambda. Our final model combined subjective and objective features with the top 10 selected interactions to achieve our best *lambda* while having a 64% accuracy, 56% balanced accuracy for Class 1, and 29% sensitivity for prediabetes.

Part II:

To analyze the predictors of diabetes status, we implemented a decision tree classification model using the **rpart** algorithm in R. We randomly split our cleaned BRFSS data into 80 percent training and 20 percent using the **createDataPartition** function, ensuring a stratified split based on the target variable **Diabetes_012**. This variable was then converted to properly treat it as a categorical outcome, representing three levels of diabetes progression: *Class 0, Class 1,* and *Class 2*. By building on the training data with an intentionally small complexity parameter (**cp = 10^-4**), we allowed the tree to explore many splits. We capped the tree depth at 5 with a minimum of 20 observations per node. This setup ensured the tree had sufficient detail in its splits, preventing overfitting. After the tree was trained, the model was pruned using the optimal cp value, which minimized cross-validation error, ensuring a more generalized tree. The final pruned tree was visualized using the **rpart.plot** function, showing key decision nodes and highlighting which variables were most influential in determining diabetes status.
Model performance was then evaluated on the test data by generating predictions and comparing them to the actual values using a confusion matrix. This matrix provided critical performance metrics, such as accuracy, sensitivity, specificity, and positive predictive value (PPV) for each diabetes class. The confusion matrix revealed that the model performed well in predicting non-diabetic individuals with high sensitivity (97.91%) but struggled with *Class 1*, showing zero sensitivity for this class. Although the model accurately identified non-diabetics, it was not as effective in detecting pre-diabetics, suggesting the need for additional or more refined features to improve prediabetes predictions. Additionally, the visualization of

the confusion matrix, both as a heatmap using base R and a polished **ggplot2** version, provided an intuitive comparison of predicted versus actual diabetes statuses. These visualizations helped identify systematic misclassifications by the model, particularly the poor performance in predicting the pre-diabetic class. The heatmaps made it easy to observe where the model confused non-diabetics and diabetics while missing pre-diabetic cases altogether. This insight suggests that the model could benefit from incorporating other variables or advanced techniques, such as ensemble methods or logistic regression models, to improve prediction for all diabetes stages. In addition to the decision tree model, a logistic regression model was implemented by collapsing the three-level diabetes indicator into a binary outcome (**Diag01**), where a value of 1 indicated a diagnosis of diabetes and 0 revealed no diagnosis. The logistic regression was run using the same predictors, and the signed coefficient estimates were analyzed. The five most significant predictors of diabetes in this model were found to be whether the individual had ever had their cholesterol checked (**CholCheck**), a history of high blood pressure (**HighBP**), heavy alcohol consumption (**HvyAlcoholConsump**), a diagnosis of high cholesterol (**HighChol**), and self-reported general health (**GenHlth**). These predictors were highly statistically significant, with individuals who had their cholesterol checked being 4.4 times more likely to report diabetes and those with high blood pressure having roughly double the odds of diabetes compared to others. Interestingly, heavy alcohol consumption was found to be negatively correlated with diabetes, suggesting a complex relationship between lifestyle factors and health outcomes.

These analyses highlight that behavioral patterns and self-reported health variables play a significant role in predicting diabetes status. While the decision tree model showed promise in predicting non-diabetic individuals, it struggled with prediabetes, suggesting that more refined or additional variables might be necessary. Further exploration using other modeling techniques, such as logistic regression or ensemble models, could help improve predictions for pre-diabetic individuals and refine the understanding of diabetes risk factors. Next, we collapsed our three-level diabetes indicator into a simple 0/1 outcome (**Diag01** = 1 if the respondent had diagnosed diabetes, otherwise 0) and ran a standard logistic regression on the same predictors. We examined and ranked the signed coefficient estimates (omitting the intercept). The five strongest predictors by absolute log-odds were whether someone has ever had their cholesterol checked (**CholCheck**), a history of high blood pressure (**HighBP**), heavy alcohol consumption (**HvyAlcoholConsump**), a diagnosis of high cholesterol (**HighChol**), and self-rated general health (**GenHlth**). Individuals who ever had their cholesterol checked were about 4.4 times more likely to report a diabetes diagnosis than those who hadn't, making cholesterol screening the most powerful predictor in our model.. **HighBP** carried roughly twice the odds of diabetes, and HighChol had about a 1.7-fold increase. Interestingly, heavy drinkers showed only about half the odds of diabetes (perhaps reflecting lifestyle or reporting differences rather than a real protective effect). Finally, poorer general health ratings were linked to about a 1.6-fold rise in diabetes risk. These five factors were highly statistically significant, underscoring their real influence on our model.

Next, we ran a logistic regression using our five top predictors and converted the model's coefficients (and their 95% confidence intervals) into odds ratios. We then visualized these with ggplot2. The odds-ratio plot reveals that cholesterol screening stands out as the strongest risk factor: individuals who had ever had their cholesterol checked were more than four times as

likely to report a diabetes diagnosis (OR ≈ 4.4; 95% CI [3.5, 6.0]). A history of high blood pressure doubled the odds of diabetes (OR ≈ 2.0), and those diagnosed with high cholesterol had about a 1.7-fold increase in risk. Self-rated poor general health carried an odds ratio of roughly 1.6. Interestingly, heavy alcohol consumption was associated with lower reported diabetes (OR ≈ 0.5), suggesting that frequent heavy drinkers were about half as likely to report diabetes—though this result may reflect reporting bias or other unmeasured lifestyle factors.

Conclusion

The goal of this project is to answer real life questions to better the health of Americans, specifically in relation to diabetes, which is a growing crisis in this country. Our questions were: We attempted to answer these questions through analysis and modeling using the data from the CDC & Behavioral Risk Factor Surveillance System (BRFSS) 2021 phone survey on Diabetes.

Reflecting on our methodology, our project primarily used techniques taught in STOR320, but we slightly extended our modeling by incorporating Elastic Net regularization. We also tried methodologies beyond the scope of our syllabus, but we found that the methods within course material perform better in answering our research questions. Our research questions aimed to apply course skills towards an innovative and integrative understanding of current health datasets. Overall, while our data was relatively clean and standard, we faced moderate analytical challenges. First, severe class imbalance, especially the low prevalence of diabetic individuals, have impaired model sensitivity. Even after upsampling, sensitivity for prediabetes (Class 1) remained low (about 29% sensitivity in the best model). In terms of implication, this may weaken the usefulness of the screening tool in real-world triage settings and calls for methods to penalize misclassification of prediabetes more heavily. From that, to enhance model robustness and prediction accuracy, it also calls for implementing ensemble methods like Random Forest, KNN, and even stacking models. Although some of these models (e.g. KNN) demonstrated insufficiency in this dataset, while doing trials and errors in web-scraping and other outside resources, we have found relative success in certain methods. Thirdly, one drawback of the dataset we chose is a reliance on self-reported survey data. This could introduce potential recall bias, social desirability bias(e.g. Underreporting smoking, overreporting exercise), and inaccuracies in self-assessment (e.g. GenHlth). This may lead to a reduced internal validity. Thus, we encourage future research to test our best model on different datasets (e.g. another year. Extending beyond that, our dataset also lacks generalizability as the dataset is restricted to a limited scope of population, and future expansion in other cultural, racial, and age cohorts is highly encouraged. Here, although mentioned in Elastic Net, we would suggest future models to more systematically engineer the interactional features, such as the non-linear interactions between complex relationships of social factors (e.g. insurance status x income x BMI). This could help better understand the compounding influences among different factors across physical and behavioral landscapes, therefore closely locate factor modeling on individual levels. Lastly, BRFSS is a cross-sectional survey, not longitudinal, which cannot predict diabetes development over time, only association at a single point. Nevertheless, the study still created valuable findings that yield relevant real-life implications.

From the modeling of the first question, we discovered that AnyHealthcare and NoDocCost were the two predictors with the most significance within the self-reported and behavioral variables in predicting someone's diabetes status. Using the model that is only based on self-reported and behavioral variables, we got a 64% accuracy in prediction of diabetes status. This should be of concern to law markers and those at the head of communities. The United States does not have universal healthcare, which

unfortunately often results in people not receiving the medical attention that they need due to cost of care. If something non medical like not having insurance coverage or not going to the doctor because of the cost is such an important predictor of diabetes, there may be a solution to this lending cause of death that does not require medical breakthroughs. For the second question, our analysis of a decision tree and logistic regression using all the variables attempted to tell us which health indicators were most useful in predicting diabetes. The decision tree performed well in predicting people without diabetes, it was not successful in identifying prediabetic individuals. This could be due to the small number of people who answered the survey who reported having prediabetes. The logistic regression model, which was trained on a binary version of the data (either diabetes or no diabetes) highlighted five important health indicators in predicting diabetes. The strongest risk factor was whether someone had a cholesterol check or not. This does not seem an obvious predictor, but it leads to more thoughtful consideration of our data.

This issue of the lack of people reporting that they have prediabetes may be due to people not knowing themselves whether or not they have prediabetes. If some of the most meaningful predictors of diabetes revolve around a lack of medical care, how can we expect the most at risk people to know that they are at risk? Diabetes is an expensive disease, requiring insulin, medical supplies and technology, and specialist visits. We hope that our project can help raise awareness for the fact that this disease is more complex than initially suspected. It is not easily predicted by vegetable consumption or BMI. There are deep social determinants that come into play when producing predictions for who may be subject to diabetes.

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